

Virtual Tours with Accessibility Annotations: A Prototype for Campus Accessibility Assessment for Mobility-Disabled Students

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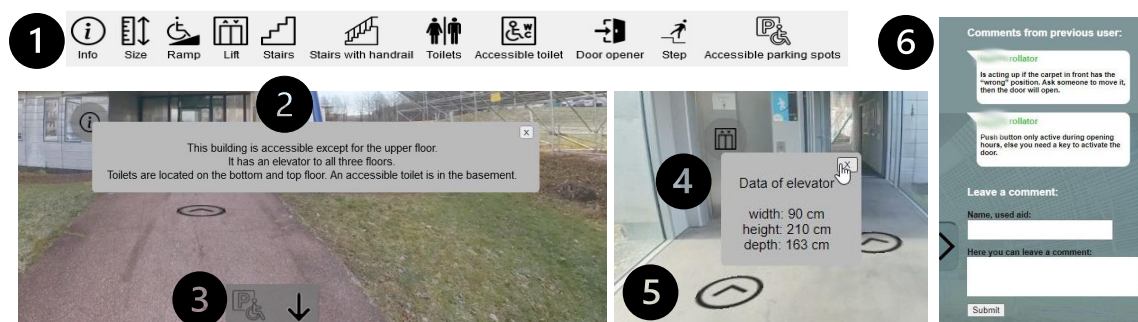


Figure 1: Screenshots of our prototype's core functionalities: (1) icons legend of the depicted information, (2) accessibility overview, (3) guiding sign icon, (4) textual details, (5) navigation arrow, (6) comment sidebar.

ABSTRACT

Mobility-disabled students often face barriers when visiting campus. Consequently, they rely on detailed planning to meet their individual needs. The planning process depends on accessibility information, especially of building interiors. This project examines the potential of a virtual tour featuring 360-degree images with accessibility annotations. A proof-of-concept prototype was developed based on requirements gathered through a focus group. The resulting prototype enables users to virtually explore the building to judge accessibility. Our initial findings suggest that combining 360-degree imagery with textual information could enable more accurate accessibility judgments tailored to individual needs than existing information sources.

Index Terms: Virtual tours, accessibility annotations, accessibility assessment, mobility disability.

1 INTRODUCTION

Despite compliance with accessibility regulations, access barriers can persist on university campuses [4]. In particular, the access to classrooms, laboratories and other campus facilities is often non-trivial [3]. Thus, to plan visits to unfamiliar environments, such as lecture halls, it is essential for people with mobility disabilities to assess accessibility using up-to-date accessibility information, ensuring their access needs are met [8]. Previous work [9] has already shown that searching for information on websites remotely is a common practice. In doing so, individuals hope to find accessibility-related descriptions, photos, and videos. Online platforms like Google Maps¹ are almost always used in this process. Another helpful option is to call and ask contact persons for verbal

descriptions, pictures, and videos, or to question friends and family [8, 9].

Virtual tours represent real places as virtual environments, aiming to let users experience a destination in advance. They are already widely used in the tourism and real estate domains [7]. Virtual replicas of indoor environments have been used for accessibility assessment [2] and embodied exploration [9]. Further, virtual tours for university campuses already exist, e.g. for campuses in Germany [10] and the U.S. [5] among others; however, to the best of our knowledge, they do not include detailed accessibility information. Our research thus aims to bridge recent work on remote accessibility assessment [2, 3, 6, 9] with a virtual campus tour to assist university students in planning visits to lecture halls and navigating the campus. Therefore, we first assessed information needs and provided a proof-of-concept implementation of one university building.

2 METHOD

To develop and evaluate our prototype, we did two initial exploratory studies. The first was a focus group with two mobility-disabled students (P1 and P2) and one assistant (P1a), to assess the current availability of campus accessibility information and to elicit requirements for the prototype, in particular, the information needed to visualise. P1 used an electric wheelchair, and P2 occasionally uses a walking stick. In the final study to evaluate the prototype, online interviews were conducted with four participants, including P1, P1a, and P2. The goal was to evaluate the implemented prototype and explore the potential of virtual tours in accessibility assessment. The study was structured into three parts. For the first two parts, participants were instructed to plan a visit to a university building using only existing sources, and to plan a visit to the same building using our prototype. While doing so, they were asked to share their screens. Finally, they were asked to rate the prototype.

3 REQUIREMENTS

According to our workshop participants, the available campus maps are often confusing and difficult to read, with missing information about lifts and stairs. The participants desire accurate, intuitive digital maps or information sources, which are easy to locate and

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¹<https://www.google.de/maps>

understand. Further, participants stress the individuality of access needs. According to participants, locations are frequently labelled as 'accessible' or 'not accessible'. P2 critiques: "I always find it very difficult when it says 'not accessible'. It usually means it's not accessible with a wheelchair, but what does that mean for me? Are there a vast number of steps or are there only three steps [...]".

We analyzed the discussions of our participants [1] and the following six requirements resulted: (1) **shown imagery** should include the way to the entrance, the entrance itself and the interior (2) **camera perspective** should allow overview and zooming in; (3) **highlight important aspects** as points of interests (POI), barriers and access features; (4) **enable navigation** to POI; (5) **provide textual details** such as gradients, size details of doors or elevators, stair characteristics and (6) **comment functionality** to provide experiences and missing information.

4 PROTOTYPE AND EVALUATION

As a proof-of-concept, a prototype tour of a single building was implemented. 39 360-degree images were taken with an Insta360 X3 camera², to obtain sufficient views from all rooms. These images were then the basis for creating a virtual tour as a web application using *Three.js*³, *Node.js*⁴ and *Vite*⁵. The prototype features a zoom and rotate function, navigation arrows and icons from *Flaticon*⁶ to enhance the tour with key information of a building as well as barriers. There are interactive icons (round) that provide additional information on click and non-interactive icons, which function as guiding signs (rectangular). Core functionalities are illustrated in Fig. 1.

When evaluating existing information sources, participants found that much information is available online; however, it is scattered across multiple sources. All participants thought that the virtual tour extended the information they found online. P2 and P3 generally appreciated the detailed information given throughout the tour, such as door width, ramp gradients and lift size. With the virtual tour, participants could individually assess the location using information not readily apparent in existing sources. P1 was able to estimate distances and sizes and P3 appreciated the details on door pushes that was incorrectly presented in existing sources. Throughout the virtual tour, all participants evaluated the comment sidebar as helpful. P3 found it useful to detect conflicts and missing information in existing sources. P2 saw it as a means to enhance the data quality of the information. However, participants wished for more details on the camera angle to better judge how objects appear in reality (P2, P3) and an integrated map of the building layout (P3).

5 CONCLUSION AND FUTURE WORK

This project explored the annotation of 360-degree images within a virtual tour in the context of campus accessibility. Through a requirement analysis, the needs of students with mobility disability were collected. Based on these requirements, a proof-of-concept prototype was developed and evaluated. The initial evaluation showed that the combination of visual and textual information provides a more comprehensive overview of accessibility barriers

and accommodations. In cases where the visual perspective is insufficient to determine accessibility, additional textual annotations facilitate judgment. Further, we also noted that accessibility is individual and merely providing information regarding compliance to standards might not be enough, in line with Hamraie [4]. Consequently, our prototype enables the documentation and judgment of accessibility information on an individual level, with detailed facts and the opportunity to share experiences.

Still, the presented prototype and initial evaluation are a proof of concept and provide room for future exploration. The prototype should be further evaluated with more students to gain even more insights into the individual requirements of students with mobility disabilities and beyond. Additionally, our prototype needs to be extended to the entire campus in the future and ideally embedded within university websites. Further, obtaining reliable and up-to-date accessibility information is the backbone of our proposed campus tour. To obtain such data at scale, artificial intelligence (AI) approaches, such as the proposed accessibility assessment of 2D images by Huang et al. [6], could also be investigated for 360-degree content.

We hope that our concept of an annotated virtual tour inspires future research, both in terms of virtual environments to show as well as the underlying data collection and assessment. We saw initial tendencies that our prototype could fill gaps of existing information sources, such as websites, and thus facilitate the planning process. Thus, annotated virtual tours should be investigated further as a means to provide accessibility information, potentially lowering the hurdle to accessing unknown locations.

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²<https://www.insta360.com/product/insta360-x3>

³<https://threejs.org/>

⁴<https://nodejs.org/en>

⁵<https://vite.dev/>

⁶Sources on flaticon.com :

Icons by Stockio: [Info](#);

Icons by Smartline: [Size](#);

Icons by Smashicon: [Ramp](#), [Accessible parking sports](#);

Icons by Freepik: [Navigation arrow](#), [Lift](#), [Stairs](#), [Accessible WC](#), [Toilet](#);

Icons by Cursor Creative: [Stairs with Handrail](#),

Icons by Ayub Irawan: [Door Opener](#);

Icons by Roundicon: [Guiding Arrow](#),

Icons by Artifex: [Step](#)